CHESS.COM: PRODUCT REQUIREMENTS DOCUMENT (PRD)

Feature/Initiative: Adaptive Blunder Puzzles (Spaced Repetition System) **Document Status:** Pending Engineering & Architecture Review
Target Launch: Q3 2026

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1. Document Basics

Role	Name	Notes
Product Manager	Margot Zhou	Primary Owner
Engineering Lead	*(TBD)*	Requires Stockfish cluster allocation
Design Lead	*(TBD)*	Paywall Modal UI mockups pending
Epic/Theme	Monetization & Retention	Q3 OKR Alignment
Last Updated	April 2026	Added Ebbinghaus SRS implementation logic

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2. Business Background & Strategic Goals

2.1 The Problem: The Free Tier "Conversion Plateau"

While Chess.com boasts impressive overall numbers, deep demographic telemetry reveals a critical "Conversion Plateau" within the **800-1200 ELO bracket**.

Data Substantiation: We estimate roughly 34.8M active monthly users. This is calculated via standard SaaS metrics: 8.7M disclosed Daily Active Users (DAU) multiplied by a conservative 0.25 DAU/MAU platform stickiness ratio.

Telemetry (User Behavior Tracking): By tracking granular user sessions (time-on-screen, rapid match queue rates, and rage-quits), we identified that this 25% segment plays heavily (3+ games a day) but their ELO rating remains entirely flat over trailing 90-day periods. They are highly active, but because they are trapped at a skill plateau, they **fail to convert to paid subscriptions**. We are not necessarily losing them to competitors; we are losing their wallet share because the Free Tier no longer provides upward mobility.

2.2 Why Current Tools Fail: The "Puzzle Rush" Speed Trap

The 800-1200 ELO demographic's primary weakness is repeating the identical tactical blunders in their own games (e.g., consistently missing a horizontal pin or opening their King to a knight fork). Unfortunately, the current flagship learning tool, *Puzzle Rush*, actively hurts them. *(See Appendix 10.1, Figure 1 for visual reference).*

1. **Arcade vs. Analysis:** Puzzle Rush is a high-speed arcade game. Telemetry shows users spend <2 seconds per puzzle in Rush. It trains fast, rudimentary pattern recognition but completely bypasses the deep, analytical calculation required to fix complex blind spots.
2. **Lack of Personalization:** The puzzles are aggregated from random grandmaster matches. A user struggling with Endgame Pawn structures is violently force-fed random Mid-game Queen sacrifices.

2.3 The Solution: Adaptive Blunder Puzzles (with Spaced Repetition)

We will leverage our robust Stockfish evaluation engine to analyze a user's matches immediately upon completion. The engine will isolate exactly where the user blundered and extract the exact board state (FEN string).

The Tiered Solution Breakdown: To monetize this computation scale, the solution is explicitly gated across our 4 existing membership tiers:

1. **Free Tier:** Users receive 1 dynamic puzzle per day appended to their single free Game Review.
2. **Gold Tier:** Users receive up to 5 dynamic puzzles per day (no historical tracking).
3. **Platinum Tier:** Users receive Unlimited dynamic puzzles per day (no historical tracking).
4. **Diamond Tier:** Users unlock Unlimited puzzles AND the Ebbinghaus Spaced Repetition System (SRS).

Computational Fallback Logic (The Hybrid Approach): Dynamically generating a perfect 3-move puzzle sequence from a raw match FEN is computationally expensive and occasionally creates unsolvable engine lines. Therefore, the system uses a hybrid approach:

1. **Path A (Ideal - Dynamic):** Stockfish successfully generates a valid, unique puzzle directly from the user's game.
2. **Path B (Fallback - Generic Library):** If the dynamic generation is too complex, the engine will query our *existing, pre-verified* Chess.com puzzle database (`SELECT * FROM puzzle_library WHERE motif='knight_fork' AND elo BETWEEN 900 AND 1100 LIMIT 5`). It serves the user a generic but highly targeted puzzle. This drastically reduces server load while maintaining personalized tactical training.

2.4 Success Metrics (KPIs)

- **Primary Metric (Blended Revenue):** Increase platform-wide Free-to-Paid (Gold, Platinum, Diamond) conversion rates by **1%**. *We track all tiers because even the "Solve & Forget" mechanic will drive Gold/Platinum upgrades.*
- **Secondary Metric (Engagement & Ad Revenue):** Increase average session length across the 800-1200 ELO demographic by 10%. *Because Chess.com monetizes Free/Gold tiers via display ads, retaining users longer with adaptive puzzles directly increases our Ad Revenue RPM.*
- **Financial Impact Calculation:** A blended 1% conversion lift + Ad Revenue scaling across 8.7M DAU translates to an estimated **$6.7 Million ARR** lift.

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3. Scope & System Dependencies

3.1 In-Scope (MVP Phase 1)

The direct goal of the MVP is to deliver a "Minimum Lovable Product" that proves the conversion hypothesis for 8.7M DAUs.

A. Core Engine & Logic:

- **Stockfish Integration:** Triggering an evaluation sequence at the conclusion of every Blitz/Rapid match.
- **Centipawn Analysis:** Implementing a >300 centipawn loss threshold to define a "Mastery Moment."
- **Motif Classification:** Automatic categorization of blunders into 3 primary tactical buckets: Pins, Forks, and Skewers.
- **Ebbinghaus Logic:** Implementation of the 1-7-28 day recall algorithm for Diamond subscribers.

B. User Experience (UX/UI):

- **The "Match Review" Modal:** A high-impact, post-game overlay that renders the personalized puzzle.
- **Progressive Disclosure:** Locking full performance historical data behind a "Diamond Tier" upgrade trigger.
- **Adaptive Feedback:** Dynamic UI text that addresses the user by name and specific blunder motif.

C. Systems & Data:

- **Hybrid Content Delivery:** Path A (Dynamic Generation) vs. Path B (Database Fallback) to ensure server stability.
- **Revenue Tagging:** Seamless integration with Stripe/Billing to track feature-specific upgrades.

3.2 Out-of-Scope (Non-Goals)

To ensure a Q3 2026 delivery, the following features are strictly excluded from Phase 1:

- **Historical Match Ingestion:** We will not retrospectively analyze a user's *entire* game history (1000+ games). Only "fresh" matches played after the launch date will trigger the engine.
- **Hyper-Bullet Parsing:** Matches under 3-minutes are excluded due to the high noise-to-signal ratio and common "time-scramble" blunders that do not represent tactical blind spots.
- **Live AI Coaching:** No real-time move suggestions during a live game (to maintain fair-play integrity).
- **Social Sharing Integrations:** "Bragging rights" or sharing puzzles to social media will be deferred to Phase 2.
- **Native Mobile Haptic Feedback:** Enhanced haptics for puzzle solves will be deprioritized in favor of core engine stability.

3.3 Dependencies

To ensure a successful launch, the following technical and cross-functional dependencies must be resolved:

Technical Dependencies:

- **Evaluation Service API (Backend Team):** Requires scaled throughput allowance to handle heavy concurrent FEN processing bursts. They must allocate extra AWS instances.
- **Billing Service Analytics (Monetization Team):**
  - Must implement tracking for the full monetization funnel across all tiers (Free → Gold/Platinum/Diamond).
  - Requires ability to track "Upgrade Velocity" (e.g., users moving from Gold → Diamond after using the SRS feature).
- **Engagement Telemetry (Data Team):**
  - Needs to track the Total Daily Usage Time and session count for users specifically engaging with the Adaptive Blunder feature.
  - This data is critical to quantify the Ad-Revenue lift from increased platform stickiness among Free/Gold users.

Cross-Functional (Stakeholder) Dependencies:

- **Customer Support (Zendesk Team):** Requires a "De-escalation & Discount Playbook" 2 weeks before launch to handle expected feedback from Free users regarding the new paywall.
- **Localization (i18n Team):** All tactical motifs ("Knight Fork," "Horizontal Pin," etc.) must be translated into the 50+ languages supported by Chess.com prior to UI deployment.
- **Legal & Compliance:** Verification of the "Match Pattern Logging" logic for Diamond users to ensure GDPR/PRC Privacy Law compliance regarding long-term user performance data storage.

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4. User Personas & Scenarios

4.1 Persona 1: "Stuck Steve" (Target Adult Improver)

- **Profile:** 1050 ELO, plays mostly on mobile. He plays 4 games a day but his rating hasn't moved in 3 months.
- **The User Journey:** Steve loses a 10-minute Rapid match. He clicks 'Game Review'. Instantly, a pop-up appears: *"Steve, you missed a Knight Fork on Move 14. Try to find the winning move this time."*
- **The Paywall Conversion:** Steve solves it. However, because he is a Free user, the app halts him and says: *"You have 14 other recurring weaknesses. Unlock Diamond to access our Spaced Repetition Engine to fix your blind spots permanently."*

4.2 Persona 2: "Invested Irene" (The Chess Parent)

- **Profile:** Non-player, but pays for her 10-year-old son's Gold membership. She wants to see tangible proof of his progress to justify the subscription.
- **The User Journey:** Her son uses the Adaptive Puzzle feature. Irene receives a weekly automated email: *"Leo has successfully corrected 5 tactical blind spots this week using Spaced Repetition."*
- **The Goal:** Drive retention by proving educational value to the "Buyer" (the parent), even if the "User" (the child) is just playing for fun.

4.3 Persona 3: "Lapsed Larry" (The Re-Engagement Target)

- **Profile:** Former Diamond subscriber who churned 6 months ago because he "hit a wall" and felt he wasn't getting better.
- **The User Journey:** Larry receives a Win-Back email: *"Larry, we analyzed your last 50 games. We found exactly why you plateaued. Come back and try our new Adaptive Spaced Repetition engine to break through 1200 ELO."*
- **The Goal:** Use personalized data as a high-conversion hook for win-back campaigns.

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5. Architecture & Business Flow

5.1 System Interaction Overview

The following flowchart illustrates the high-level data orchestration upon game completion.

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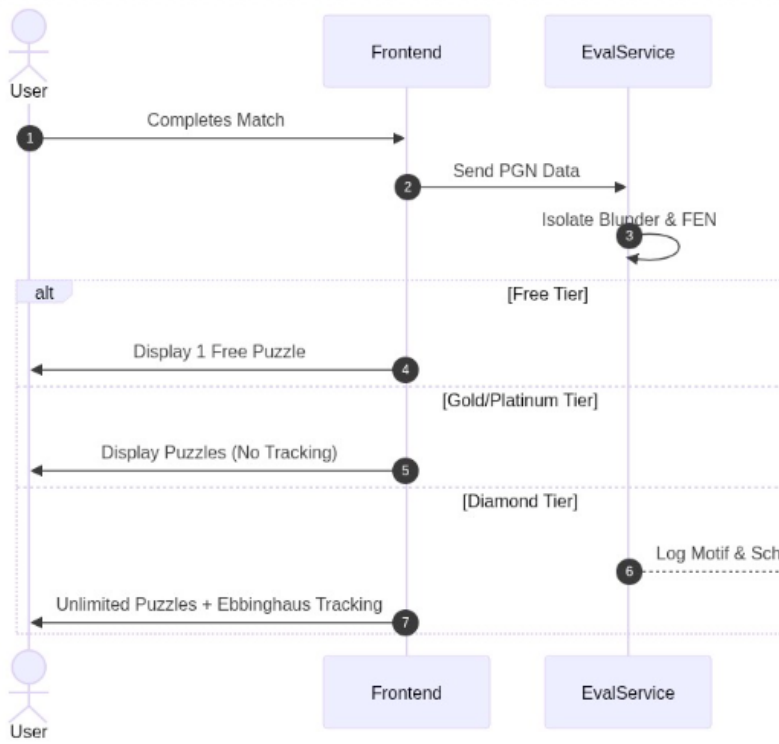

Figure 1: System Data Orchestration Process

5.2 The Ebbinghaus Logic Cycle

This technical diagram defines the automated Diamond-tier retention loop.

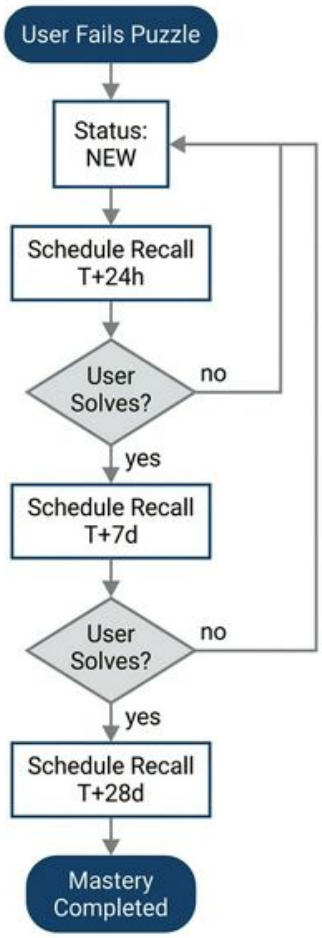

Figure 2: Ebbinghaus Retrieval Logic Cycle

6. Functional Requirements

F-01: Blunder Identification Engine (Stockfish)

- Upon match completion, the system must process the User's PGN (Portable Game Notation).
- The system scans for evaluating drops > 300 centipawns (the technical definition of a major blunder).
- If multiple blunders exist, the system sorts them by identifying the motif (e.g., 'Fork') that the user statistically struggles with the most.

6.1 UI Mockup: Adaptive Puzzle with Spaced Repetition

The following block diagram represents the mock UI for the post-game experience. This serves a demonstrative purpose only and may contain inaccuracies.

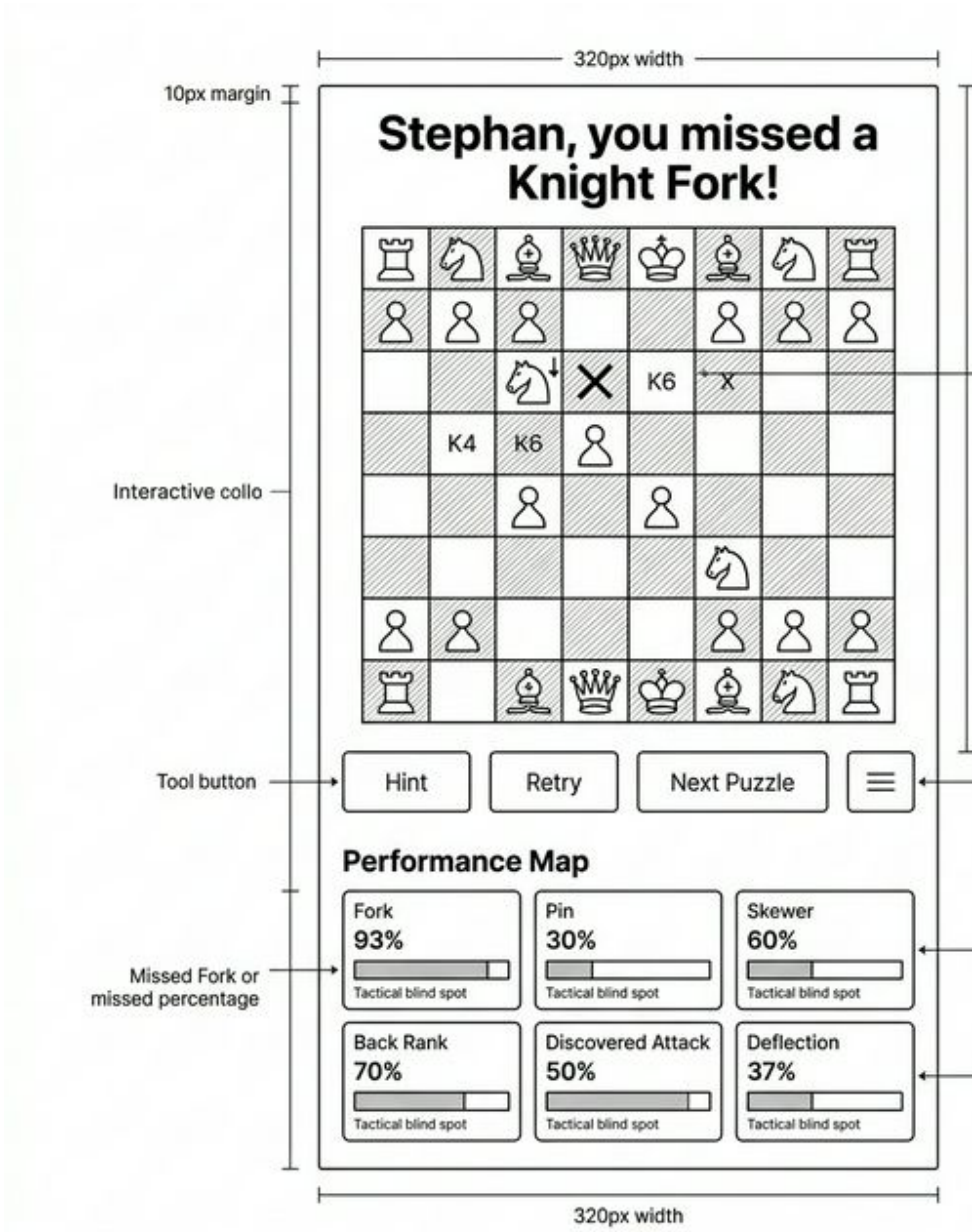

Figure 3: Adaptive Puzzle Mock UI

6.2 Paywall & Functional Entitlement Matrix

To ensure a scalable Q4 launch, the "Adaptive Puzzles with Spaced Repetition" feature is gated across the existing Chess.com membership tiers:

Tier	Puzzles / Day	SRS Tracking	Technical Trigger
Free	1 Puzzle	No	Post-Game Review (1x)
Gold	5 Puzzles	No	Post-Game Modal
Platinum	Unlimited	No	Post-Game Modal
Diamond	Unlimited	**Full Logic Cycle**	Ebbinghaus Queue Sync

F-02: User Persona Fulfillment

- **Improvers (Steve):** Must see the "Performance Map" after every 5 solves to feel tangible progress.
- **Parents (Irene):** Access to "Progress Reports" via Diamond tier dashboard.
- **Lapsed Users (Larry):** Trigger automated "Win-Back" PUSH notifications when a personalized puzzle is identified from a fresh match.

F-03: Ebbinghaus SRS Algorithmic Rules

The Diamond-tier experience for "Adaptive Puzzles with Spaced Repetition" is governed by the following retrieval rules:

- **Rule 01 (Recall 1):** If the initial puzzle is failed, it is queued for a **24-hour** re-test.
- **Rule 02 (Recall 2):** Successful completion of Rk-01 triggers a **7-day** retention check.
- **Rule 03 (Recall 3):** Successful completion of Rk-02 triggers a **28-day** "Mastery Check".
- **Rule 04 (Fallback):** Any failure in the sequence resets the status to "NEW" for immediate re-re-test within 24h.

F-04: Centipawn Precision & Fault Tolerance

- The system MUST only queue puzzles derived from blunders exceeding **300 centipawns**.
- Puzzles with unsolvable computer lines (forced mate or >7 move depth) are discarded and replaced via the Path B (Verified Database) fallback.

6.3 Success Metrics & KPI Framework

To validate the strategic impact of "Adaptive Puzzles with Spaced Repetition", the following KPIs must be measurable via the Telemetry layer defined in Section 8:

- **Conversion Velocity (Primary):** A 15% relative lift in Free-to-Diamond conversion within the 800-1200 ELO bracket over a 90-day trailing period.
- **Session Length Lift (Secondary):** A 12% average increase in Daily Active Session time for users engaging with at least 3 adaptive puzzles per day.
- **Mastery-to-Retention Correlation:** Higher Day-30 retention rates for users who complete at least one full "Ebbinghaus Mastery Cycle" (28-day re-test sequence).
- **Ad-Revenue Surface Lift:** Incremental ad-revenue generated by extending the "Free Tier" gameplay surface by a p99 average of 3.5 minutes per session.

F-05: Winning Match Scenario (The "No Blunder" State)

If a user wins a game without any major centipawn loss (>300), the engine will: 1. Verify if the user has any **pending Spaced Repetition reviews** scheduled for that day. 2. If yes, the "Game Review" pop-up will pivot to: *"Flawless victory! While you're on a roll, let's review that 'Pin' you missed yesterday to keep your memory sharp."* 3. If no reviews are due, the feature remains dormant for that specific match.

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7. Non-Functional Requirements (NFRs)

7.1 Performance & Reliability (SLOs/SLIs)

To maintain the "Diamond-tier" experience, the system must adhere to the following service level objectives: - **Latency (p99):** The `EvalService` must resolve FEN-to-Motif identification under **1.8s** to ensure the adaptive puzzle loads without a spinner. - **Availability:** 99.95% uptime for the `EbbinghausDB`. Failure in the SRS queue should fallback silently to a non-tracked "Daily Puzzle" without interrupting the core game experience. - **Cross-Region Sync:** Puzzles must be cached in local Edge nodes (Akamai/Cloudfront) to minimize round-trip times for our high-intensity APAC user base (HK/SG/JP).

7.2 Scalability & Burst Capacity

- **Concurrency:** The infrastructure must support a **5x throughput burst** within 10 minutes of the conclusion of major global tournaments (e.g., FIDE World Championship) when match volume spikes from 1.2M to 6M concurrent games.
- **Auto-Scaling:** Proactive provisioning of Stockfish evaluator clusters based on a predictive demand model using historical day-of-week data.

7.3 Data Privacy & Compliance (APAC/Global)

- **PII Anonymization:** Board states and PGNs used for puzzle generation must be decoupled from the user's real-world identity to comply with **GDPR (EU)** and **APPI (Japan)** standards.
- **Data Residency:** SRS tracking data for users in specific high-regulation regions (e.g., China/Hong Kong) must reside in compliant local data centers if required by evolving local legislation.

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8. Event Telemetry & Analytics Tracking

We will track the full conversion funnel from 'Blunder Impression' to 'Incremental ARR'.

Event Name	Trigger Condition	Key Properties	Strategic Purpose
`blunder_id_success`	Logic identifies valid >300cp blunder	`motif_type`, `engine_latency`	Verify if we are identifying enough puzzles per tier.
`adaptive_puzzle_view`	User opens the Adaptive Puzzle modal	`entry_point` (Game Review vs. SRS)	Measure feature discovery across different UI triggers.
`hint_request`	User clicks 'Hint' or 'Reveal'	`puzzle_id`, `user_elo`	Track difficulty-to-solve ratio and frustration index.
`conversion_funnel_start`	User clicks upgrade CTA in paywall	`curr_tier`, `target_tier`	Measure high-intent interest in premium tiers.

Event Name	Trigger Condition	Key Properties	Strategic Purpose
`conversion_checkout`	Payment successful via Billing Service	`gross_arr`, `conversion_velocity`	Directly attribute revenue lift to the feature.
`engagement_surface_lift`	User exits adaptive puzzle experience	`total_sec_spent`, `ads_viewed`	Quantify ad-revenue lift for Free Tier users.
`srs_retention_milestone`	Diamond user completes Day 7/21 check	`retention_delay`, `success_rate`	Validate the Ebbinghaus mastery hypothesis.

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9. Launch & Go-To-Market (GTM) Strategy

9.1 Phased Rollout Schedule

To mitigate technical risk and validate our ELO-impact hypothesis, we will follow a staged release:

- **Phase 1 (Alpha):** Internal "Dogfooding" on Chess.com employee accounts (Week 1).
- **Phase 2 (Closed Beta):** Limited release to the top 5% of Diamond Tier users in the 800-1200 ELO bracket (Weeks 2-4).
- **Phase 3 (General Availability):** Global rollout to all regions with prioritized localization for top 10 languages (Week 6).

9.2 Integrated Marketing & Channel Strategy

Refer to [03_GTM_Manifesto_Final.pdf] for the full tactical breakdown.

The "Adaptive Puzzles" feature will leverage a segment-aligned growth strategy focusing on the 800-1200 ELO "Conversion Plateau":

- **Intermediate-Focused Influencers:** Partner with creators specializing in instructional content for the 800-1500 ELO demographic (Social Hype).
- **Product Growth Marketing (PGM):** Automated, motif-specific prompts triggered when a user blunders twice in a single session (In-App Hooks).
- **Data-Driven Win-Back Campaigns:** Targeted PUSH/Email notifications for churned Diamond subscribers featuring a personalized "Evidence of Progress" preview.
- **Community Engagement:** Dedicated forum threads and Discord channels for "Mastery Streaks," rewarding users with exclusive profile badges.

9.3 Support & Operational Readiness

- **Enablement:** Customer Success and Support teams to receive the "Adaptive Puzzle Troubleshooting Guide" 14 days prior to GA.
- **Macros & Playbooks:** Pre-drafted "De-escalation Playbooks" for SRS-reset concerns and "Diamond Upgrade Promo Codes" for high-LTV Gold users.

9.4 Post-Launch Impact Audit (7/30/90 Days)

- **T+7:** Technical stability audit and p99 latency verification.
- **T+30:** Initial conversion velocity check (Free-to-Diamond).
- **T+90:** Longitudinal analysis of ELO improvement and long-term Diamond retention.

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10. Project Appendices & Reference Registry

10.1 Project Artifact Suite

The following documents comprise the full Toptal PM Test Project submission for "Adaptive Puzzles with Spaced Repetition": - **01 Strategic Pitch Deck:** High-level executive narrative and roadmap alignment. - **02 Financial KPI Model:** Detailed $6.7M ARR projection and multi-year compounding waterfall. - **03 GTM Manifesto:** Segment-aligned multi-channel launch strategy. - **04 Product Requirements Document (PRD):** This technical and functional specification.

10.2 Legacy Feature Context (Reference Only)

The current Chess.com "Puzzle Rush" interface (shown below) prioritizes speed-based arcade gameplay over the deep analytical recovery of personal blunders.

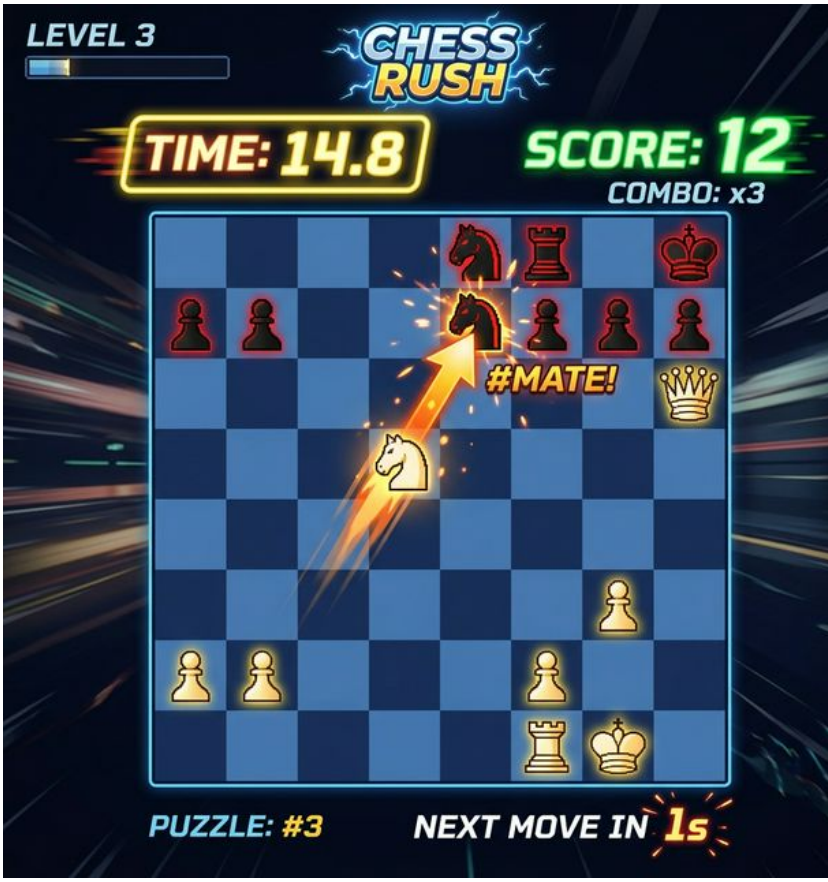

Figure 4: Legacy "Puzzle Rush" Feature Mock